# PR. 2 — Data Cleanser

## Theory Concepts & Definitions

Missing Value Imputation and Outlier Handling for Healthcare Data Preprocessing

## 1. Missing Value Imputation

### 1.1 Simple Imputer

Replaces every missing value in a column with a single summary statistic computed from the observed (non-missing) values of that column. For numerical columns the statistic is typically the **mean** or **median**; for categorical columns it is the **most frequent value (mode)**. It is the simplest and fastest imputation technique, but because it inserts the same repeated value into every gap, it artificially reduces the column's variance and can distort its distribution shape.

### 1.2 Most Frequent Imputation

A special case of Simple Imputer used for categorical data: every missing entry is replaced with the category that occurs most often in the column. It is easy to apply and keeps the column fully valid (no new categories introduced), but it can over-represent the majority class if a large share of values were missing.

### 1.3 Missing Indicator + Random Sample Imputation

A two-part technique. First, a **Missing Indicator** creates a new binary (0/1) column that flags exactly which rows originally had a missing value in a given column — this preserves the possibility that *missingness itself* is informative (e.g. a lab test wasn't ordered). Second, **Random Sample Imputation** fills each gap by drawing a real, randomly-selected value from the column's own observed data (with replacement), rather than inserting one fixed number. This keeps the post-imputation distribution's shape and variance much closer to the original than mean/median imputation does.

### 1.4 KNN Imputer (k-Nearest Neighbors)

A multivariate imputation method. For a row with a missing value, KNN Imputer finds the *k* most similar rows (by distance across the other numeric columns) and fills the gap using a (distance-weighted) average of those neighbors' values in the target column. Because it uses relationships between variables rather than a single column statistic, it typically produces more realistic estimates — at the cost of higher computation time and sensitivity to unscaled features or unresolved outliers.

### 1.5 MICE (Multivariate Imputation by Chained Equations)

The most sophisticated method covered here (implemented as **IterativeImputer** in scikit-learn). Each column with missing values is modeled as a regression problem, predicted from all other columns. The algorithm cycles through every column repeatedly ("chained equations"), refining the imputed values each pass until they stabilize. MICE generally gives the least-biased estimates for data that is Missing At Random (MAR), since it explicitly captures the conditional relationships between variables, but it is also the slowest and least transparent method to explain.

# 2. Outlier Detection and Treatment

## 2.1 Z-score Method

Measures how many standard deviations a value sits from its column's mean: $z = (x - mean) / std$. A common threshold flags any point with $|z| > 3$ as an outlier, since under a roughly normal distribution only about 0.3% of points should naturally fall that far out. It is simple and fast, but assumes the data is close to normally distributed, and the mean/std it relies on can themselves be skewed by the very outliers it's trying to detect.

## 2.2 IQR (Interquartile Range) Method

Measures spread using the middle 50% of the data: $IQR = Q3 - Q1$ (the difference between the 75th and 25th percentiles). Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged as an outlier. Because it is based on quantiles rather than the mean, it is a more "robust" statistic — much less sensitive to the outliers themselves than the Z-score method, and generally preferred for skewed distributions. This is the same rule that determines the whiskers on a standard boxplot.

## 2.3 Percentile Method (Percentile Capping)

Sets fixed cutoffs at chosen percentiles of the data — commonly the 1st and 99th percentile — and **caps (clips)** any value beyond those bounds to the cutoff itself, rather than deleting the row. This guarantees a predictable, reproducible proportion of the data sits at each tail regardless of the column's underlying shape.

## 2.4 Winsorization

Conceptually similar to percentile capping: extreme values beyond a chosen percentile threshold (e.g. the bottom and top 1%) are replaced with the nearest boundary value rather than removed. The key advantage over deleting outlier rows is that the full sample size is preserved — no data (and no information contained in a row's other columns) is thrown away, which is especially valuable for smaller or clinically important datasets.

# 3. Method Comparison Summary

| Category | Method | Best Suited For | Key Tradeoff |
|---|---|---|---|
| Missing Values | Simple Imputer (mean/median/mode) | Quick baseline, low-missingness columns | Reduces variance; distorts distribution shape |
| | Most Frequent | Categorical columns | Over-represents majority class |
| | Missing Indicator + Random Sample | When missingness itself may be informative | Preserves distribution shape well; adds extra columns |
| | KNN Imputer | Numeric data with correlated features | More accurate; computationally heavier |
| | MICE | Data Missing At Random (MAR) with multiple correlated columns | Most accurate; slowest, least transparent |

| Category | Method | Best Suited For | Key Tradeoff |
|---|---|---|---|
| Outliers | Z-score | Roughly normal distributions | Sensitive to the outliers it detects |
| | IQR | Skewed distributions (robust) | Fixed multiplier (1.5) may not fit every column |
| | Percentile Capping | Predictable, reproducible bounds | Not adaptive to actual distribution shape |
| | Winsorization | Preserving full sample size | Still shrinks true extreme values toward center |

Note: This document summarizes the theory behind the techniques applied in *Data_Cleanser.ipynb*. All definitions and comparisons reflect standard data-preprocessing practice as used in the accompanying notebook.